ADEMOLA AFOLABI COLLINS

Full Stack Engineer

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PROFESSIONAL SUMMARY

Backend & Full-Stack Engineer with 3 years of hands-on experience building REST APIs, transactional systems, and database-backed applications with TypeScript, Node.js, PHP, Laravel, MySQL, PostgreSQL and MongoDB. Strong focus on deterministic business logic, data integrity, authentication, concurrency control, and automated testing. Experienced in taking backend features from data modeling and API design through validation, testing, containerization, and frontend integration.

TECHNICAL SKILLS

Backend: Node.js, Express, PHP, Laravel, REST APIs, OOP, SOLID Principles
Languages: TypeScript, JavaScript, PHP, SQL
Databases: MySQL, MongoDB, PostgreSQL, Database Indexing, Query Optimization, Schema Design
Security & Integrity: JWT, OAuth2, RBAC, Input Validation, Database Transactions, Row-Level Locking
Testing & Tools: Jest, Git/GitHub, Docker, Postman, Swagger/OpenAPI, CI/CD
Frontend: Vue, React, Angular, Next.js

PROJECTS

JobNet — Full-Stack Recruitment Platform

Angular, TypeScript, PHP, MySQL, Firebase Auth, Docker, Swagger/OpenAPI

- Built **61 REST API endpoints** supporting authentication, recruitment workflows, user management, and three role-based application areas.
- Designed MySQL-backed data relationships and server-side validation for **Job Seeker, Employer, and Admin** workflows.
- Integrated Firebase authentication with the backend and implemented authorization boundaries around protected API operations.
- Containerized the backend with Docker and documented the API using Swagger/OpenAPI.

Vaultly — Digital Banking System

Vue, Pinia, Laravel, MySQL, Sanctum, Docker

- Built **21 REST API endpoints** covering deposits, withdrawals, transfers, beneficiaries, authentication, and account management.
- Designed transfer flows around **database transactions and row-level locking (SELECT ... FOR UPDATE)** to prevent conflicting concurrent balance updates.
- Modeled transactional data and optimized database queries and indexes for account and transaction-history operations.
- Tested concurrent transfer scenarios and failure paths to verify transactional behavior and data consistency.

AI-Powered Customer Support Refund System

Vue, Express, TypeScript, PostgreSQL, LLM API, Docker, Automated Testing

- Built an AI-assisted refund workflow where **deterministic TypeScript rules** validate delivery windows, item eligibility, refund limits, and other business constraints before an AI response is used.
- Separated business-rule enforcement from the LLM layer and validated structured transaction context before it reached downstream processing.
- Added safeguards for malformed and potentially malicious inputs so untrusted AI/user content could not determine refund eligibility directly.
- Built a **58-test automated suite** covering invalid inputs, expired refund windows, boundary conditions, and malicious prompts; containerized the system with Docker.

NairaNest — Banking Application

React, Redux Toolkit, Node.js, MongoDB, Material UI, JWT

- Built REST API flows for authentication, account operations, and money transfers using Node.js, MongoDB, and JWT.
- Implemented **multi-document MongoDB transactions** so sender balances, recipient records, and transaction records were updated atomically.
- Added protected routes and input validation around financial operations and authentication flows.
- Integrated backend services with the React client through structured API payloads and explicit asynchronous loading/error states.

ACHIEVEMENTS

- **100+ REST API endpoints** built across four full-stack systems, including transactional financial workflows and role-based applications.
- Completed a **6-level SQI Software Engineering program** covering Laravel, MySQL, JavaScript, MERN, Angular, Vue, Cloud Operations, and DSA.

EDUCATION

B.Tech, Computer Engineering — Ladoke Akintola University of Technology (LAUTECH)